Business Question and Visualization Report

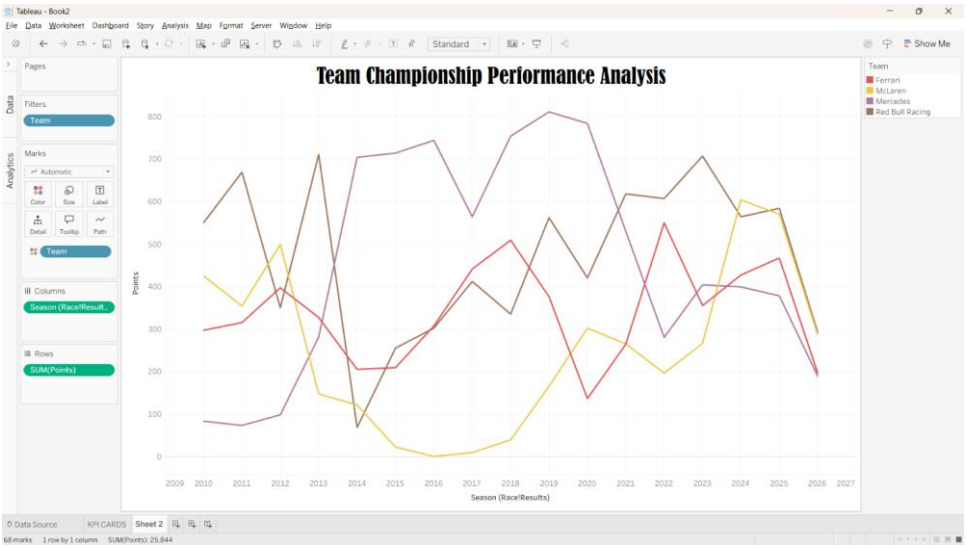
Date	2 November 2026
Team ID	PNT2022TMIDxxxxxx
Project Name	F1 Racing Performance Analysis
Maximum Marks	5 Marks

This report defines the business questions behind the F1 racing analysis in Tableau. Each question is paired with the visualization built to answer it, a screenshot of the worksheet, and the key insight it provides.

Business Questions and Visualisation

1. Which teams perform best across seasons, and how has the balance of power shifted?

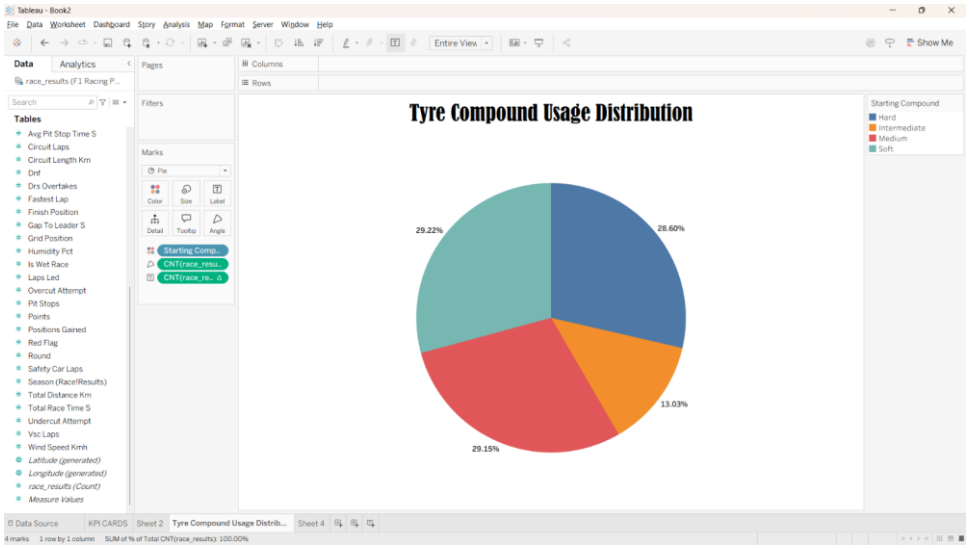
Visualization: *Line chart of total points per season for Ferrari, McLaren, Mercedes and Red Bull Racing (Team Championship Performance Analysis).*



Insight: Mercedes dominates the mid-2010s, staying above roughly 700 points from 2014 to 2020 apart from a dip in 2017, and peaks near 810 in 2019. Red Bull leads in 2011 and 2013 and again from 2021 to 2023, while McLaren climbs to about 600 points in 2024 after near-zero totals around 2016. The sharp fall for every team in 2026 probably reflects a season that is not yet complete.

2. How are starting tyre compounds distributed across all races?

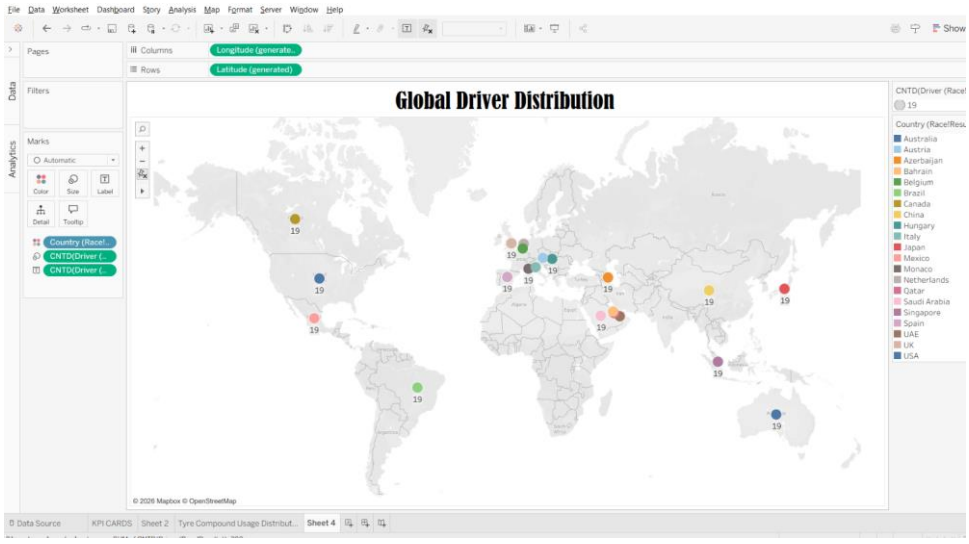
Visualization: *Pie chart showing the share of races started on each compound (Tyre Compound Usage Distribution).*



Insight: Soft (29.22%), Medium (29.15%) and Hard (28.60%) are used almost equally, while Intermediate accounts for only 13.03%. This suggests teams choose dry compounds as a balanced strategy and that wet-weather tyres are needed in a minority of races.

3. Where do the drivers in the dataset race around the world?

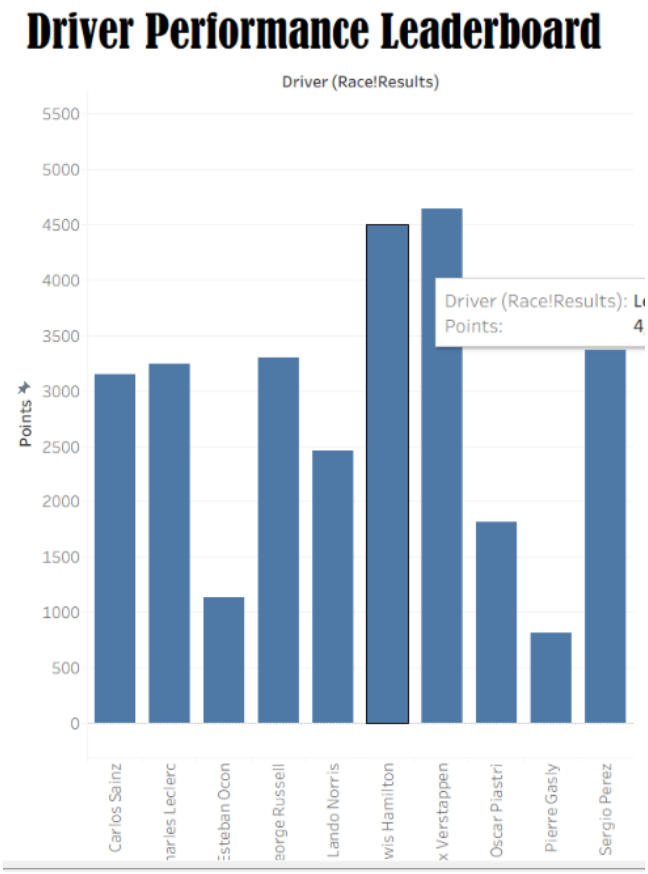
Visualization: *Map with one marker per country, coloured by country and labelled with the distinct driver count (Global Driver Distribution).*



Insight: Every marked country shows the same count of 19 distinct drivers, across 24 countries on all continents with a race. Because the count is uniform, the country field appears to describe where the race was held, not where drivers come from.

4. Who are the top-scoring drivers?

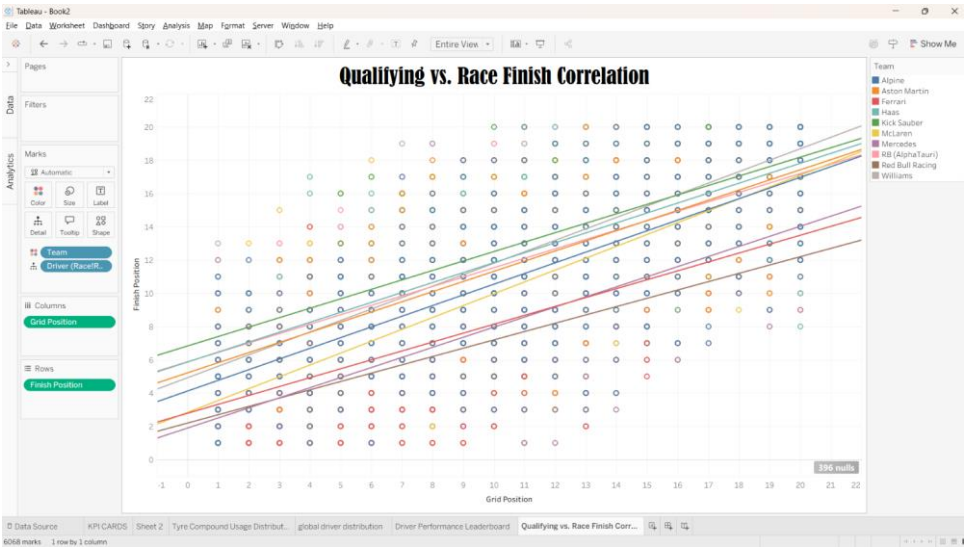
Visualization: *Bar chart of total championship points per driver (Driver Performance Leaderboard).*



Insight: Max Verstappen leads with about 4,650 points, followed closely by Lewis Hamilton with 4,491. Sergio Perez, George Russell, Charles Leclerc and Carlos Sainz form a middle group at roughly 3,100 to 3,400 points, and Pierre Gasly has the lowest total at around 800.

5. Does qualifying position predict the race finishing position?

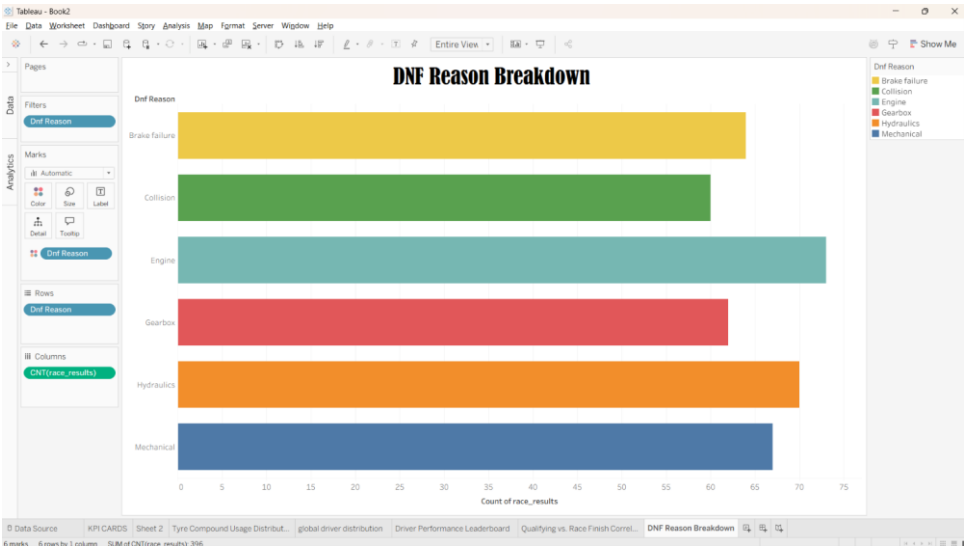
Visualization: *Scatter plot of grid position against finish position, coloured by team with a trend line per team (Qualifying vs. Race Finish Correlation).*



Insight: Every team shows a clear upward trend: starting further back generally means finishing further back. The Red Bull and Mercedes trend lines sit lowest, meaning better finishes from the same grid slot, while teams such as Williams, Kick Sauber and Haas sit highest. The chart notes 396 nulls, which are most likely retirements with no classified finish.

6. What are the main reasons drivers fail to finish a race?

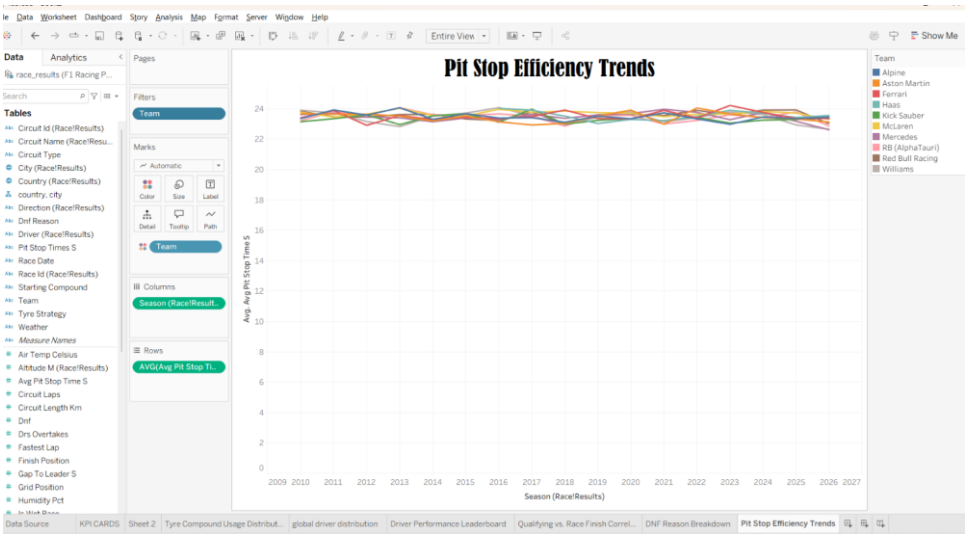
Visualization: *Horizontal bar chart of DNF counts by reason (DNF Reason Breakdown).*



Insight: Engine failure is the most common cause at about 73 retirements, followed by hydraulics (about 70) and mechanical issues (about 67). Brake failure, gearbox problems and collisions each account for roughly 60 to 64. The six reasons are close together, and reliability faults together outweigh collisions.

7. Are pit stops becoming more efficient over time?

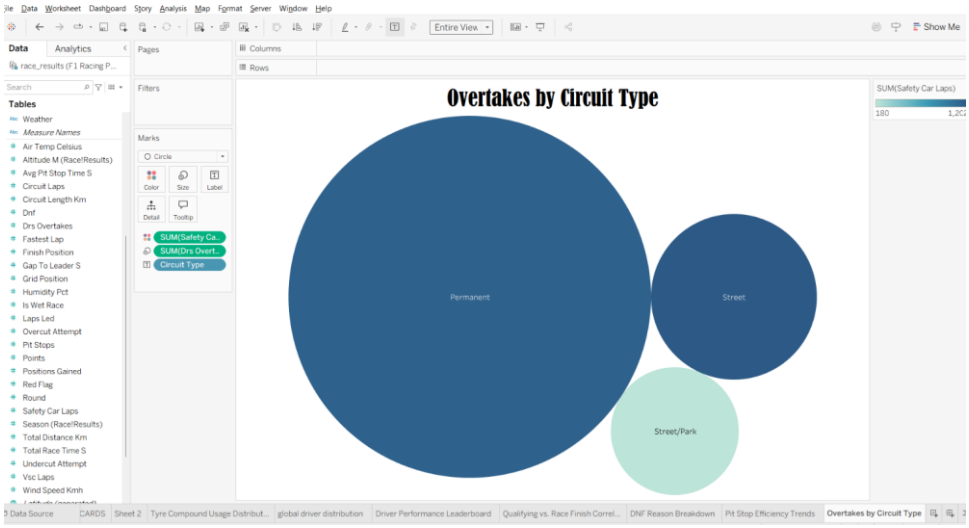
Visualization: *Line chart of average pit stop time per season for each team (Pit Stop Efficiency Trends).*



Insight: Average pit stop time stays close to 23 seconds for every team in every season, with only small variation between teams. There is no clear efficiency trend, which suggests pit stop performance is already very consistent across the grid.

8. Which circuit types produce the most overtaking and safety car activity?

Visualization: *Bubble chart where size shows DRS overtakes and colour shows safety car laps, by circuit type (Overtakes by Circuit Type).*



Insight: Permanent circuits have by far the most DRS overtakes, followed by Street circuits, with Street/Park circuits lowest. Colour shows that safety car laps also rise with overtakes, from 180 up to about 1,200 laps. Permanent circuits host the most races, so the totals partly reflect volume.

Tableau Public link

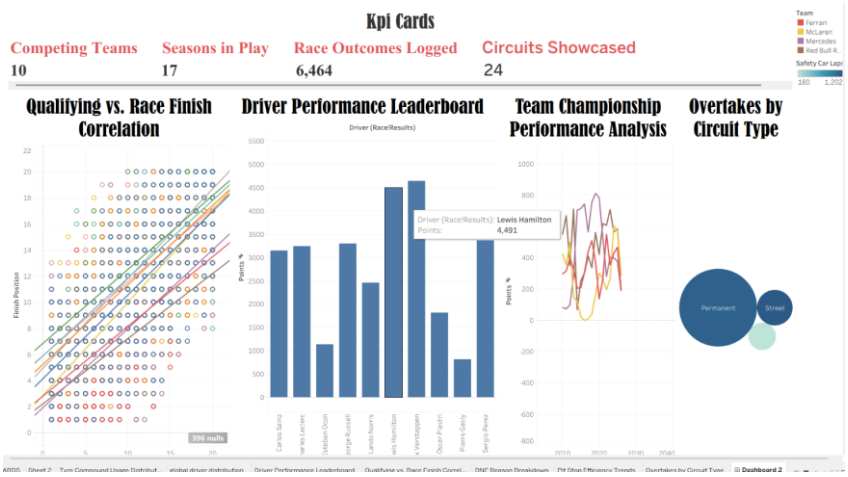
https://public.tableau.com/app/profile/hardik.sachdeva8040/viz/Dashboard1_17935473753590/Dashboard2?publish=yes

Dashboard Design

Date	2 November 2026
Team ID	PNT2022TMIDxxxxxx
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The dashboard brings the most important views together on one screen. A KPI strip at the top gives the scale of the dataset, and four charts below answer the main performance questions: grid-to-finish correlation, driver points, team trends over seasons and overtakes by circuit type. Team colours stay consistent between charts, and hovering over a bar shows the exact driver and points value.

Activity 1: Interactive and visually appealing dashboards



Major outcomes from the dashboard:

- **Dataset scale:** The KPI cards show 10 competing teams, 17 seasons, 6,464 race outcomes logged and 24 circuits, which gives a broad base for the analysis.
- **Grid position matters:** The scatter plot shows a clear positive relationship between starting and finishing position for every team. Red Bull and Mercedes convert grid slots into better finishes than the rest.
- **Top drivers:** Max Verstappen (about 4,650 points) and Lewis Hamilton (4,491 points) lead the leaderboard, well ahead of Sergio Perez, George Russell, Charles Leclerc and Carlos Sainz.
- **Shifting team dominance:** Mercedes peaks at about 810 points in 2019, Red Bull leads in 2011, 2013 and 2021 to 2023, and McLaren rises sharply in 2024.
- **Circuit type and racing action:** Permanent circuits account for the most overtakes, followed by Street and Street/Park circuits, and these also carry the most safety car laps.

Activity 2

Publish the dashboard on Tableau Public and paste the public link below:

https://public.tableau.com/app/profile/hardik.sachdeva8040/viz/Dashboard1_17935473753590/Dashboard2?publish=yes

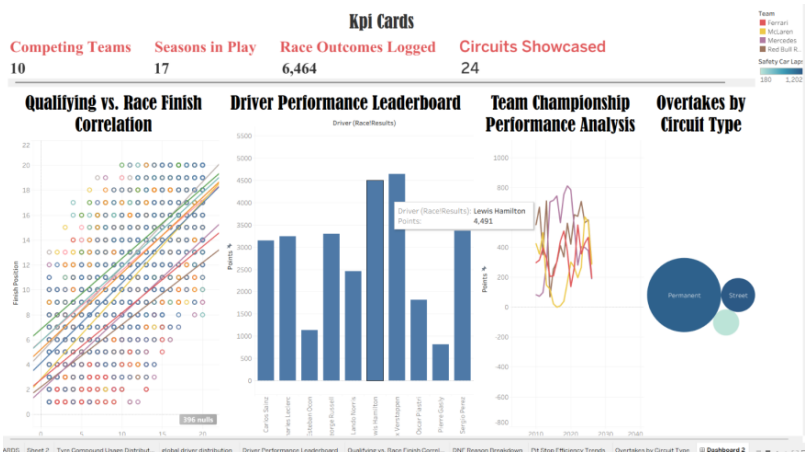
Story

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A Tableau story presents the analysis as a sequence of story points, each carrying a short caption and a supporting visual. This story takes the viewer from the scale of the data, through team and driver performance, to race strategy, reliability and circuit behaviour, and ends on the combined dashboard.

Story points

#	Caption	Visual
1	Four KPIs set the scale: 10 teams, 17 seasons, 6,464 race outcomes, 24 circuits	KPI Cards
2	Team dominance shifts: Mercedes peaks in 2019, Red Bull and McLaren surge later	Team Championship Performance Analysis
3	Verstappen and Hamilton lead the driver points leaderboard	Driver Performance Leaderboard
4	Qualifying well pays off: grid position predicts race finish for every team	Qualifying vs. Race Finish Correlation
5	Engine failures are the leading cause of DNFs	DNF Reason Breakdown
6	Permanent circuits deliver the most overtakes	Overtakes by Circuit Type
7	The full picture on one dashboard	Dashboard 2



Observations:

- **Team dominance moves in eras:** Mercedes leads through the mid-2010s and peaks near 810 points in 2019, while Red Bull and McLaren take over later in the period.

- **Starting position drives results:** Finish position rises steadily with grid position for every team, so qualifying is a strong predictor of race outcome.
- **Two drivers lead the points table:** Max Verstappen and Lewis Hamilton are clearly ahead of the rest of the leaderboard.
- **Reliability still decides races:** Engine failure is the top DNF reason, with hydraulics and mechanical faults close behind.
- **Circuit type shapes the action:** Permanent circuits produce the most overtakes and the most safety car laps.

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